

PORT SCANNING CHALLENGE ETHICAL HACKING – ASSESSMENT TOOL 2

Course Name: Ethical Hacking

Course Code: ITA14

Course Outcome: CO3

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. Experiment 1 – Nmap Port Scanning

Aim

To perform a security assessment on server 192.168.1.20, identify open ports and running services, analyze their purpose, and recommend measures to reduce the attack surface.

Command

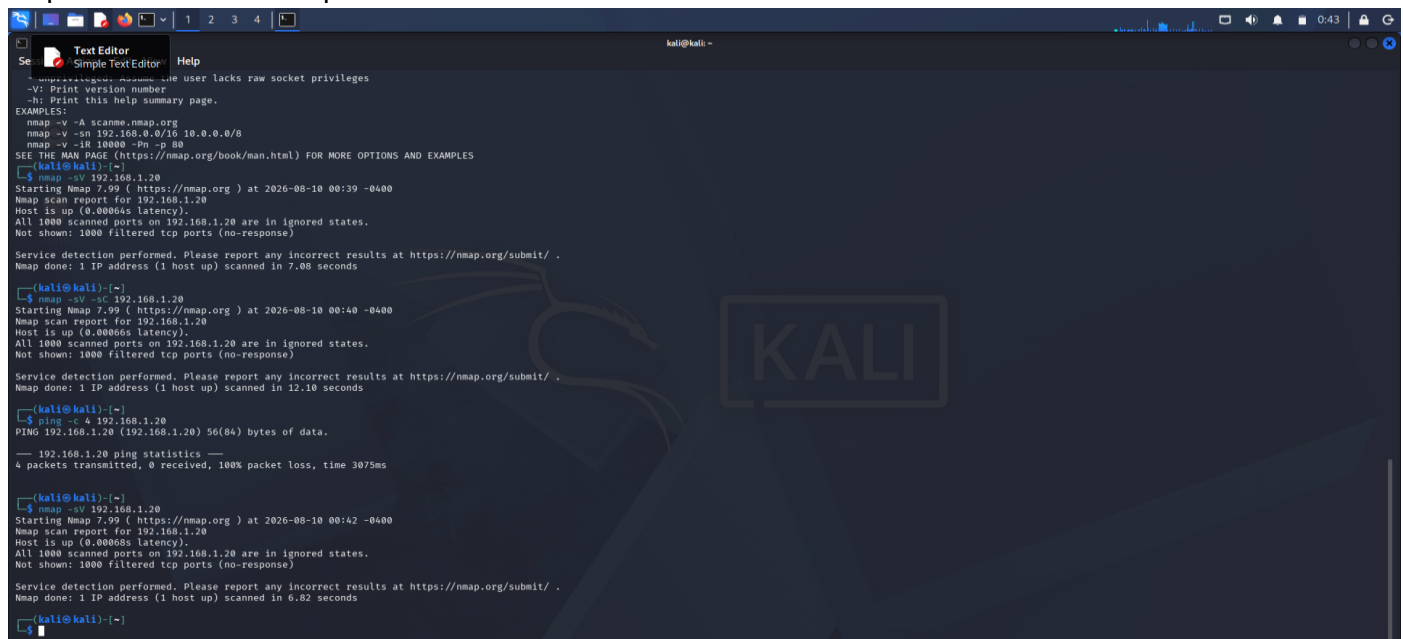
```
nmap -sV 192.168.1.20
```

Detailed scan (if permitted in the lab):

```
nmap -sV -sC 192.168.1.20
```

Procedure

1. Open the Kali Linux terminal.
2. Run the Nmap service/version detection command against the specified target.
3. Record the open ports and services shown in the output.
4. Analyze the purpose and security relevance of each discovered service.
5. Capture the terminal output as a screenshot.



```
Se Text Editor Simple Text Editor Help
-----
-V: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -pn -o 00
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
(kali@kali)-[~]
$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:39 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00066s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.08 seconds
(kali@kali)-[~]
$ nmap -sV -sC 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:40 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00066s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.10 seconds
(kali@kali)-[~]
$ ping -c 4 192.168.1.20
PING 192.168.1.20 (192.168.1.20) 56(86) bytes of data:
--- 192.168.1.20 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3075ms

(kali@kali)-[~]
$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:42 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00066s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.82 seconds
(kali@kali)-[~]
$
```

Observation

Record the actual open ports, service names, and versions displayed in the screenshot. Do not add ports that were not discovered.

Result

The Nmap assessment was performed on 192.168.1.20 and the discovered ports and services were documented for security analysis.

Experiment 2 – Nikto Web Server Assessment

Aim

To assess the web server hosted on 192.168.1.30 and identify potential web-server misconfigurations, outdated components, or exposed files and directories.

Commands

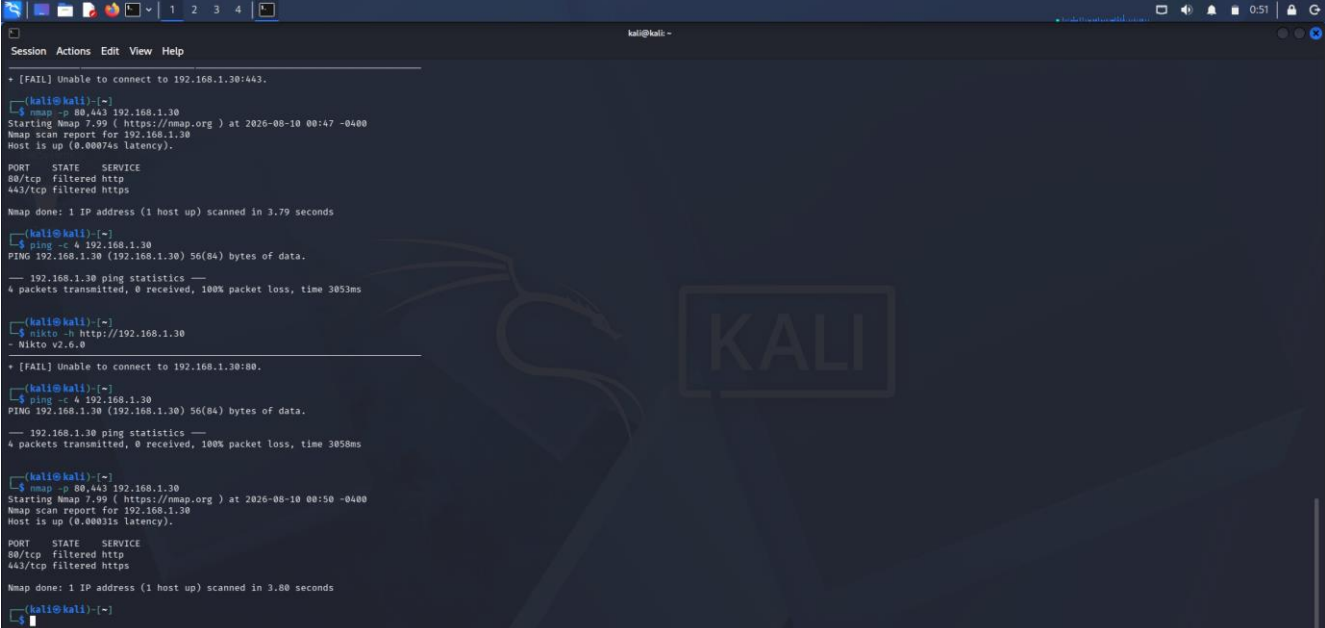
```
nikto -h http://192.168.1.30
```

For an HTTPS service, if available:

```
nikto -h https://192.168.1.30
```

Procedure

6. Check whether the target host is reachable.
7. Check whether HTTP or HTTPS is accessible.
8. Run Nikto against the available web service.
9. Record the findings and their security implications.
10. Capture the Nikto output as a screenshot.



```
kali@kali: ~  
[FAIL] Unable to connect to 192.168.1.30:443.  
  
(kali@kali)~  
$ nmap -p 80,443 192.168.1.30  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:47 -0400  
Nmap scan report for 192.168.1.30  
Host is up (0.00074s latency).  
  
PORT      STATE SERVICE  
80/tcp    filtered http  
443/tcp   filtered https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.79 seconds  
  
(kali@kali)~  
$ ping -c 4 192.168.1.30  
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data.  
--- 192.168.1.30 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3053ms  
  
(kali@kali)~  
$ nikto -h http://192.168.1.30  
- Nikto v2.6.0  
  
[FAIL] Unable to connect to 192.168.1.30:80.  
  
(kali@kali)~  
$ ping -c 4 192.168.1.30  
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data.  
--- 192.168.1.30 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3050ms  
  
(kali@kali)~  
$ nmap -p 80,443 192.168.1.30  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:50 -0400  
Nmap scan report for 192.168.1.30  
Host is up (0.00031s latency).  
  
PORT      STATE SERVICE  
80/tcp    filtered http  
443/tcp   filtered https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.80 seconds  
  
(kali@kali)~  
$
```

Observation

If the web service is accessible, record the actual Nikto findings here. If the target is not accessible, document the actual connection error instead of inventing findings.

Current troubleshooting observation, if applicable:

Nmap reported the target as up but ports 80 and 443 as filtered, and Nikto reported that it was unable to connect. Replace or supplement this section with the final output obtained from the lab.

Result

The web-server assessment was attempted against 192.168.1.30. The final result should be recorded from the actual Nikto output.

Experiment 3 – Netcat Banner Investigation

Aim

To investigate host 192.168.1.40 by connecting to an authorized open port and collecting available service banner information.

Commands

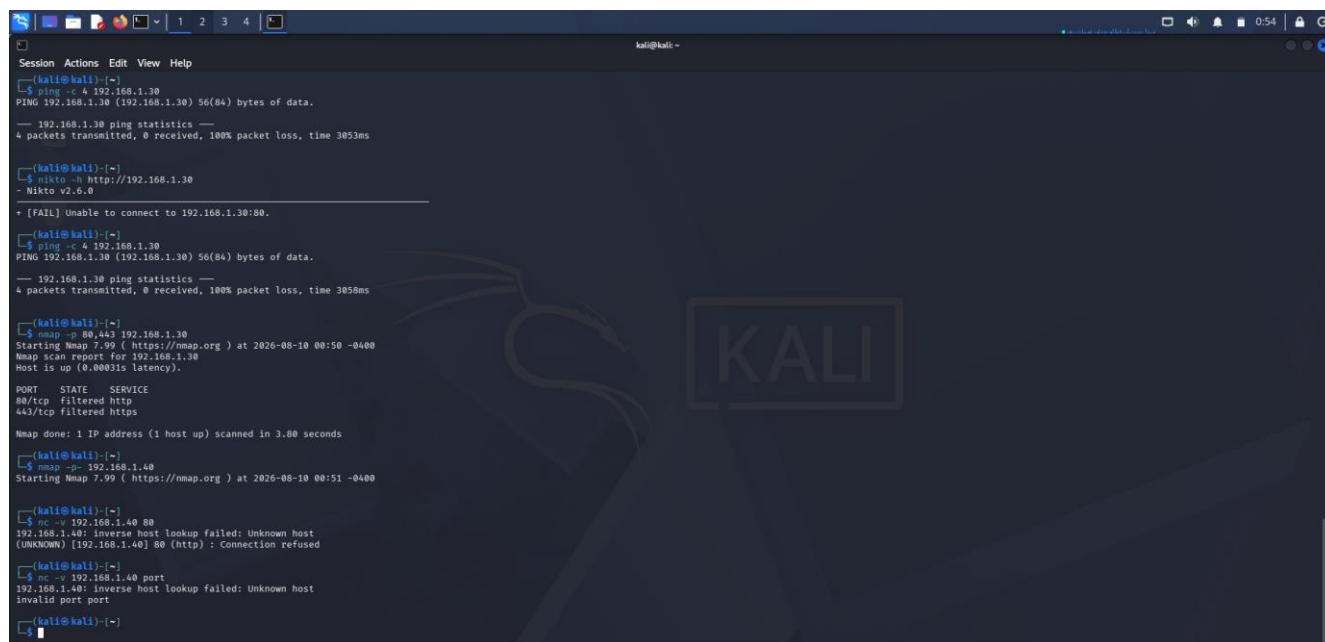
```
nmap -p- 192.168.1.40
```

```
nc -v 192.168.1.40 PORT
```

For an HTTP service: `printf "HEAD / HTTP/1.0\r\n\r\n" | nc -v 192.168.1.40 80`

Procedure

11. Identify an open service port on the authorized target.
12. Connect to the open port using Netcat.
13. Record the banner or service information returned.
14. Explain how banner information can assist reconnaissance.
15. Capture the output screenshot.



```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali:~$ ping -c 4 192.168.1.30  
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data:  
--- 192.168.1.30 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3053ms  
  
kali@kali:~$ nikto -h http://192.168.1.30  
- Nikto v2.6.0  
+ [FAIL] Unable to connect to 192.168.1.30:80.  
  
kali@kali:~$ ping -c 4 192.168.1.30  
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data:  
--- 192.168.1.30 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3058ms  
  
kali@kali:~$ nmap -p 80,443 192.168.1.30  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:50 -0400  
Nmap scan report for 192.168.1.30  
Host is up (0.00031s latency).  
  
PORT      STATE SERVICE  
80/tcp    filtered http  
443/tcp   filtered https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.80 seconds  
  
kali@kali:~$ nmap -p- 192.168.1.40  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:51 -0400  
  
kali@kali:~$ nc -v 192.168.1.40 80  
192.168.1.40: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.1.40] 80 (http) : Connection refused  
  
kali@kali:~$ nc -v 192.168.1.40 port  
192.168.1.40: inverse host lookup failed: Unknown host  
invalid port port  
  
kali@kali:~$
```

Observation

Port used: _____

Banner/service information observed: _____

Analysis

Service banners may reveal service names, versions, protocol details, or other information useful during reconnaissance. Unnecessary version disclosure should be minimized and exposed services should be kept updated.

Result

Netcat was used to investigate an authorized open service on 192.168.1.40 and the available banner information was documented.

Experiment 4 – Telnet Remote Administration Examination

Aim

To examine server 192.168.1.50 to determine whether remote administration services are accessible and identify related security concerns.

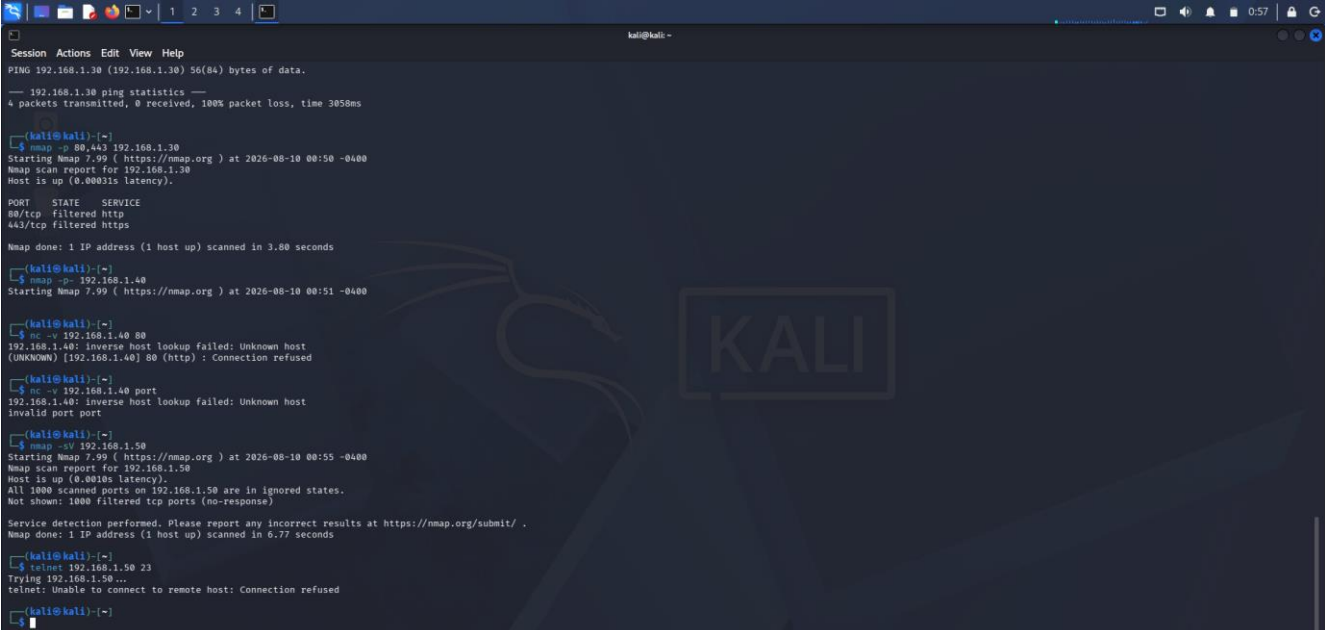
Commands

```
nmap -sV 192.168.1.50
```

```
telnet 192.168.1.50 23
```

Procedure

16. Scan the authorized server for accessible services.
17. If Telnet is available, connect to the appropriate port.
18. Record the connection result.
19. Analyze the security implications of insecure remote administration.
20. Capture the terminal output.



```
kali@kali: ~  
Session Actions Edit View Help  
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data:  
--- 192.168.1.30 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3058ms  
  
kali@kali:~$ nmap -p 80,443 192.168.1.30  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:50 -0400  
Nmap scan report for 192.168.1.30  
Host is up (0.00031s latency).  
  
PORT      STATE SERVICE  
80/tcp    filtered http  
443/tcp   filtered https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.80 seconds  
  
kali@kali:~$ nmap -p 192.168.1.40  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:51 -0400  
  
kali@kali:~$ nc -v 192.168.1.40 80  
192.168.1.40: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.1.40] 80 (http) : Connection refused  
  
kali@kali:~$ nc -v 192.168.1.40 port  
192.168.1.40: inverse host lookup failed: Unknown host  
invalid port port  
  
kali@kali:~$ nmap -sV 192.168.1.50  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:55 -0400  
Nmap scan report for 192.168.1.50  
Host is up (0.0018s latency).  
All 1000 scanned ports on 192.168.1.50 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 6.77 seconds  
  
kali@kali:~$ telnet 192.168.1.50 23  
Trying 192.168.1.50...  
telnet: Unable to connect to remote host: Connection refused  
  
kali@kali:~$
```

Observation

Accessible service/port: _____

Connection result: _____

Security Concerns

Telnet is generally considered insecure for administration because it does not provide the same encrypted protection expected from secure remote administration.

Recommended Alternative

Use SSH for secure remote administration, apply strong authentication, restrict administrative access, and keep the service updated.

Result

The remote administration accessibility of 192.168.1.50 was examined and security recommendations were documented.

Experiment 5 – Netdiscover Host Discovery

Aim

To perform host discovery on the authorized network segment 192.168.1.1–50 and analyze how discovered information may support further enumeration.

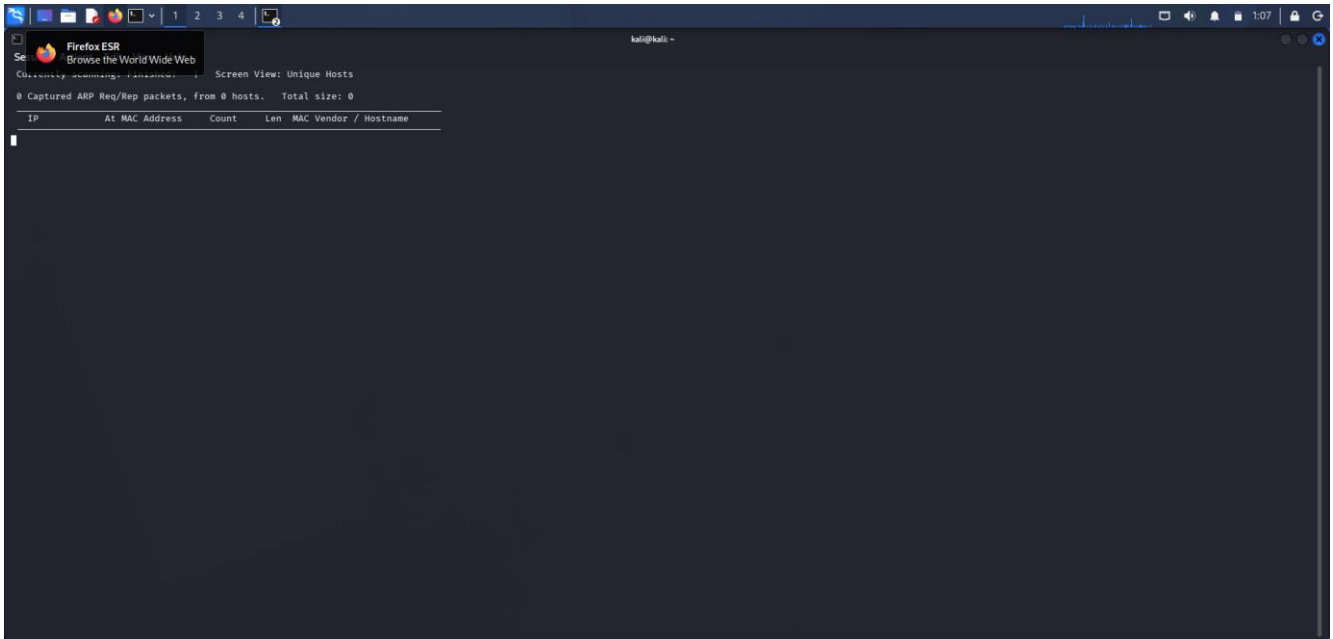
Commands

```
ip addr
```

```
sudo netdiscover -r 192.168.1.0/24
```

Procedure

21. Check the local network interface and address.
22. Run Netdiscover on the authorized lab network.
23. Identify active hosts that fall within the required address range.
24. Record IP addresses, MAC addresses, and vendor information shown.
25. Capture the discovery output.



Analysis

Host discovery can reveal which systems are active and provide information that can guide authorized enumeration. Network segmentation, access controls, firewalls, and monitoring can reduce unnecessary exposure.

Result

Netdiscover was used for host discovery on the authorized network segment and the observed active hosts were documented.

